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1. FILE OPERATIONS

```
bipin@bipin-VirtualBox:/$ ls
bin  cdrom  etc  lib  lost+found  mnt  proc  run  snap  swap.img  tmp  var
boot  dev  home  lib64  media  opt  root  sbin  srv  sys  usr

bipin@bipin-VirtualBox:/$ mkdir -/linux_lab
error: unexpected argument '-/' found

    tip: to pass '-/' as a value, use '-- -/'

Usage: mkdir [OPTION]... DIRECTORY...

For more information, try '--help'.
bipin@bipin-VirtualBox:/$ mkdir ~/linux_lab
bipin@bipin-VirtualBox:/$ cd ~/linux_lab
bipin@bipin-VirtualBox:~/linux_lab$ nano greeting.txt
bipin@bipin-VirtualBox:~/linux_lab$ cat greeting.txt
"hello linux!"
bipin@bipin-VirtualBox:~/linux_lab$ cp greeting.txt backup.txt
bipin@bipin-VirtualBox:~/linux_lab$ mv backup.txt old_greeting.txt
bipin@bipin-VirtualBox:~/linux_lab$ ls
greeting.txt  old_greeting.txt
bipin@bipin-VirtualBox:~/linux_lab$ rm old_greeting.txt
bipin@bipin-VirtualBox:~/linux_lab$ mkdir temp_dir
bipin@bipin-VirtualBox:~/linux_lab$ rmdir temp_dir
bipin@bipin-VirtualBox:~/linux_lab$ ls
greeting.txt
bipin@bipin-VirtualBox:~/linux_lab$ ls -l greeting.txt
-rw-rw-r-- 1 bipin bipin 15 Sep  8 11:32 greeting.txt
bipin@bipin-VirtualBox:~/linux_lab$
```

Figure 1: file operations using linux command

Summary

In this we used basic commands to create folders and files and used nano. We copied and rename the file and moved them. We also removed the file and directories.

2. APT Package Management

```
bipin@bipin-VirtualBox:~/linux_lab$
bipin@bipin-VirtualBox:~/linux_lab$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://np.archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://np.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://np.archive.ubuntu.com/ubuntu resolute-backports InRelease
9 packages can be upgraded. Run 'apt list --upgradable' to see them.
bipin@bipin-VirtualBox:~/linux_lab$ sudo apt install build-essential
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-30      linux-image-unsigned-7.0.0-30-generic  linux-modules-7.0.0-30-generic  linux-tools-7.0.0-30-generic
  linux-headers-7.0.0-30-generic  linux-main-modules-zfs-7.0.0-30-generic  linux-tools-7.0.0-30
Use 'sudo apt autoremove' to remove them.

Installing:
  build-essential

Installing dependencies:
  bzip2      g++      g++-x86-64-linux-gnu  libalgorithm-merge-perl  libfakeroot  lto-disabled-list
  dpkg-dev  g++-15  libalgorithm-diff-perl  libcrypt-dev            libfile-fcntllock-perl  make
  fakeroot  g++-15-x86-64-linux-gnu  libalgorithm-diff-xs-perl  libdpkg-perl            libstdc++-15-dev

Suggested packages:
  bzip2-doc  debian-keyring  debian-tag2upload-keyring  g++-multilib  g++-15-multilib  gcc-15-doc  git  bzip  libstdc++-15-doc  make-doc

Summary:
  Upgrading: 0, Installing: 18, Removing: 0, Not Upgrading: 9
  Download size: 19.3 MB
  Space needed: 73.4 MB / 13.6 GB available

Continue? [Y/n] y
Get:1 http://np.archive.ubuntu.com/ubuntu resolute/main amd64 libstdc++-15-dev amd64 15.2.0-16ubuntu1 [2,920 kB]
Get:2 http://np.archive.ubuntu.com/ubuntu resolute/main amd64 g++-15-x86-64-linux-gnu amd64 15.2.0-16ubuntu1 [14.4 MB]
Get:3 http://np.archive.ubuntu.com/ubuntu resolute/main amd64 g++-15 amd64 15.2.0-16ubuntu1 [26.7 kB]
Get:4 http://np.archive.ubuntu.com/ubuntu resolute/main amd64 g++-x86-64-linux-gnu amd64 4:15.2.0-5ubuntu1 [968 B]
Get:5 http://np.archive.ubuntu.com/ubuntu resolute/main amd64 g++ amd64 4:15.2.0-5ubuntu1 [1,100 B]
Get:6 http://np.archive.ubuntu.com/ubuntu resolute/main amd64 make amd64 4.4.1-3 [197 kB]
```

Figure 2: APT commands 1.1

```
Processing triggers for libc-bin (2.43-2ubuntu2.3)...
bipin@bipin-VirtualBox:~/linux_lab$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

bipin@bipin-VirtualBox:~/linux_lab$ apt search python3
alembic/resolute 1.18.4-1 all
  lightweight database migration tool for SQLAlchemy

antimony/resolute 0.9.3-2.1build2 amd64
  Trash 3D computer-aided design CAD tool

astro-frameworks/resolute 5.0 all
  Astronomical data analysis frameworks

astro-python3/resolute 5.0 all
  Python 3 packages for astronomy

bamkit/resolute 0.0.1+git20170413.ccd079d-3build1 all
  tools for common BAM file manipulations
```

Figure 3: APT command 1.2

Summary

In this section, we installed essential development tools on a Linux system and later verified the installation.

3. C Compilation

Step 1 : write the code

```
Standalone python3(y).dll import library generator - Rust source code
bipin@bipin-VirtualBox:~/linux_lab$ nano hello.c
bipin@bipin-VirtualBox:~/linux_lab$ cat hello.c
#include <stdio.h>

int main(){
    int a = 10, b = 5;

    printf("hello, Linux Lab!\n");
    printf("Sum of %d and %d is %d\n", a, b, a + b);

    return 0;
}
bipin@bipin-VirtualBox:~/linux_lab$
```

Figure 4: create source file

Summary

In this section a C program file named hello.c was created and inside the file some contents were written.

Step 2: Compile and Run

```
Compilation terminated.
bipin@bipin-VirtualBox:~/linux_lab$ nano hello.c
bipin@bipin-VirtualBox:~/linux_lab$ gcc hello.c -o hello_program
bipin@bipin-VirtualBox:~/linux_lab$ ls -l hello_program
-rwxrwxr-x 1 bipin bipin 15992 Sep  8 12:04 hello_program
bipin@bipin-VirtualBox:~/linux_lab$ ./hello_program
hello, Linux Lab!
Sum of 10 and 5 is 15
bipin@bipin-VirtualBox:~/linux_lab$ gcc -Wall hello.c -o hello_program_warn
bipin@bipin-VirtualBox:~/linux_lab$ ./hello_program_warn
hello, Linux Lab!
Sum of 10 and 5 is 15
bipin@bipin-VirtualBox:~/linux_lab$
```

Figure 5: compile and execute

Summary

In this step, we compile the C program using the GCC compiler. As a best practice, you can also compile the code with warnings enabled

Step 3: Modify and Recompile

```
Sum of 100 and 200 is 300
bipin@bipin-VirtualBox:~/linux_lab$ nano hello.c
bipin@bipin-VirtualBox:~/linux_lab$ gcc -Wall hello.c -o hello_program_warn
bipin@bipin-VirtualBox:~/linux_lab$ ./hello_program_warn
hello, Linux Lab!
Sum of 100 and 200 is 300
bipin@bipin-VirtualBox:~/linux_lab$
```

Figure 6: Iterative Development

Summary

In this step we only change the input numbers and recompile them and run to check the changes made.

4. Cleanup and Verification

```
Sum of 100 and 200 is 300
bipin@bipin-VirtualBox:~/linux_lab$ cd ~
bipin@bipin-VirtualBox:~$ ls -lh ~/linux_lab/
total 40K
-rw-rw-r-- 1 bipin bipin 15 Sep  8 11:32 greeting.txt
-rw-rw-r-- 1 bipin bipin 152 Sep  8 12:08 hello.c
-rwxrwxr-x 1 bipin bipin 16K Sep  8 12:04 hello_program
-rwxrwxr-x 1 bipin bipin 16K Sep  8 12:09 hello_program_warn
bipin@bipin-VirtualBox:~$ cat ~/linux_lab/hello.c
#include <stdio.h>

int main(){
    int a = 100, b = 200;

    printf("hello, Linux Lab!\n");
    printf("Sum of %d and %d is %d\n", a, b, a + b);

    return 0;
}
bipin@bipin-VirtualBox:~$
```

Figure 7: Final check

Summary

In this last step we see all our work and list all the files in lab directory with details. We also view our content of the c program file.

